

Analysis of Pellet Intake Patterns in Mice

Objective

The goal is to analyze pellet intake events in mice to distinguish between "meals" and "mega-meals" based on inter-pellet intervals (IPIs) and cluster sizes, using methods such as exponential decay and conditional probability.

1 Inter-Pellet Interval (IPI) Calculation

The IPIs measure time intervals between consecutive pellet intakes, which we later use to identify clusters.

Formula:

$$\text{IPI}_i = t_{i+1} - t_i \quad (1)$$

where:

- t_i and t_{i+1} are timestamps of consecutive pellet intake events.

Code:

```
# Calculate IPIs from timestamps  
IPIs = np.diff(np.array(pellettimes))
```

2 Kernel Density Estimation (KDE)

We visualize the distribution of IPIs using KDE to observe clustering behavior and select an IPI threshold for defining clusters.

KDE Formula:

$$\text{KDE}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right) \quad (2)$$

where:

- K is the Gaussian kernel function,
- h is the bandwidth parameter,
- n is the total number of IPI data points.

Code:

```
# KDE plot to determine threshold for clustering  
sns.kdeplot(IPIs, bw_adjust=0.5)
```

3 Clustering Pellet Events

Using a 60-second threshold, we group pellet intake into clusters. Clusters with 2-5 pellets are classified as "meals," while larger clusters are "mega-meals."

Code:

```
# Cluster pellets based on 60s IPI threshold  
clusters = cluster_pellets_by_ipi(pellettimes, ipi_threshold=60)
```

4 Exponential Decay Fitting on Cluster Sizes

To understand how frequency decreases with increasing cluster size, we fit an exponential decay model.

Formula:

$$y(x) = a \cdot e^{-b \cdot x} \quad (3)$$

where:

- $y(x)$ is the frequency of clusters with size x ,
- a represents the initial frequency,
- b is the decay rate.

Code:

```
# Fit exponential decay model to cluster sizes
params, _ = curve_fit(lambda x, a, b: a * np.exp(-b * x), sizes, counts)
```

5 Log Transformation for Linear Regression

To confirm the exponential decay, we apply a log transformation to linearize the model.

Transformation Formula:

$$\ln(y(x)) = \ln(a) - b \cdot x \quad (4)$$

Code:

```
# Apply log transformation for linear regression
log_counts = np.log(counts)
```

6 Conditional Probability of Cluster Expansion

We calculate the probability of a pellet event expanding by one pellet to identify the likelihood of cluster growth.

Conditional Probability Formula:

$$P(x+1|x) = \frac{\text{Number of } (x+1)\text{-pellet clusters}}{\text{Number of } x\text{-pellet clusters}} \quad (5)$$

Code:

```
# Calculate conditional probability of cluster growth
P_x_plus_1_given_x = np.array([(counts[i+1] / counts[i]) for i in range(len(counts)-1)])
```

7 Maximum Gap Method for Boundary Determination

To finalize the boundary between meals and mega-meals, we identify the largest gap in conditional probabilities.

Formula:

$$\theta = \max(P(x+1|x) - P(x|x-1)) \quad (6)$$

Code:

```
# Determine threshold by finding the largest gap in conditional probabilities
threshold = np.argmax(np.diff(P_x_plus_1_given_x)) + 1
```

8 Wilson Score Interval for Confidence Intervals

Using the Wilson Score Interval, we calculate confidence intervals for each conditional probability.

Wilson Score Interval Formula:

$$CI = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \quad (7)$$

where:

- $\hat{p}$ is the observed proportion,
- z is the z-score (e.g., 1.96 for 95%),
- n is the sample size.

Code:

```
# Calculate Wilson score confidence intervals
def wilson_score_interval(p, n, z=1.96):
    return (p + (z**2 / (2 * n)) - z * np.sqrt((p * (1 - p) / n) + (z**2 / (4 * n**2)))) / (1 + (z**2 /
    (p + (z**2 / (2 * n)) + z * np.sqrt((p * (1 - p) / n) + (z**2 / (4 * n**2)))) / (1 + (z**2 /

# Calculate for each conditional probability
for p, n in zip(P_x_plus_1_given_x, counts[:-1]):
    wilson_score_interval(p, n)
```

Conclusion

Through these steps, we analyzed clustering behavior of pellet intake in mice, validating the classification of 2-5 pellet clusters as meals and larger clusters as mega-meals. This process involved IPI calculation, KDE for clustering, exponential decay, conditional probability, and confidence intervals, with each method progressively refining meal boundary determination.